LIAN YANG

Python & AI Integration Developer | FastAPI · React · LLMs
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SUMMARY

Python developer who builds LLM applications end to end (RAG with cited answers, multi-stage AI pipelines, validated structured outputs) with offline tests, cost controls and secure defaults. 3 years of freelance full-stack work with React, Python and PostgreSQL. Informatics Engineering candidate (thesis stage). Seeking remote backend, full-stack or AI integration roles with US/Canada teams.

TECHNICAL SKILLS

Backend: Python, FastAPI, Pydantic, Typer, REST APIs, PostgreSQL
AI / LLM: OpenAI-compatible APIs, tool calling, structured JSON outputs, RAG (LangGraph, LangChain, FAISS, BM25 + embeddings), Whisper ASR, TTS, output validation
Frontend: React, JavaScript, Vite, HTML/CSS
Mobile: Android, Kotlin, Jetpack Compose, on-device TTS
Tools & quality: Git/GitHub, GitHub Actions, Docker, pytest with test doubles, FFmpeg, n8n

SELECTED PROJECTS

Portfolio AI Assistant | *FastAPI, LangGraph, LangChain, FAISS, React* catbearlang.github.io/portfolio

- Built a bilingual (EN/ES) recruiter assistant as a LangGraph state graph over hybrid search (BM25 + embeddings in FAISS) of the project docs; answers cite numbered sources, and a grounding check retries or declines unsupported answers.
- The same agent operates the website on request through tool calling (theme, language, sections, projects, CV download); the server validates each call against a whitelist and the browser carries it out visibly.
- FastAPI on Render with per-visitor and daily quotas and an answer cache; 47 offline tests with a fake LLM, a 32-question evaluation script and GitHub Actions CI.

ELF Reader | *Kotlin, Jetpack Compose, on-device TTS* Android

- Built an Android e-book reader that opens EPUB, Markdown, Word, TXT, text PDF and HTML and reads them aloud in the background like an audiobook.
- Voices are generated on the phone with Supertonic, an on-device TTS model, using the Snapdragon 8 Elite NPU when available: no server and no per-minute cloud cost.
- Reading extras: animated backgrounds, ambient sounds, an illustration gallery and a sleep timer.

GlosSub | *Python, Whisper, LLM, Pydantic, FastAPI* glossub.shop

- Built a six-stage cached pipeline (ffmpeg → Whisper → segmentation → LLM translation → phrase alignment → render) that produces color-aligned bilingual subtitles.
- On a real 23.7-minute episode: 99% of cues translated (253/254) and 89% phrase-alignment coverage (688 groups); re-render from cache in under 1 second.
- Shipped as a web app supporting five languages in both directions, with sentence replay, a subtitle editor and ASS/SRT export; LLM output is validated with Pydantic and semantic checks before use.

Codex Stickman Pipeline | *Python, LLM, TTS, FFmpeg, pytest* github.com/CatbearLang/codex-stickman-pipeline

- Built a resumable CLI that turns one topic into a narrated 1080p explainer video through six idempotent stages (script → voice → align → images → compose → verify).
- Designed fail-closed cost controls: hard caps on paid image calls, pre-spend estimates and a durable budget ledger, so interrupted or concurrent runs never overspend.
- Validated LLM-written scripts against a JSON schema plus pacing and word-budget gates before any paid API call.
- 94 passing offline tests using provider test doubles; built-in secret scanner; swappable LLM, TTS and alignment providers.

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer | *Remote* 2023 – Present

- Developed and maintained web applications for small-business clients, covering React frontends, backend APIs and PostgreSQL databases.
- Turned client requirements into working features, from data modeling and API design to delivery and post-launch support.
- Diagnosed and fixed issues in existing codebases, improving stability and shipping incremental feature updates.

IT Support Technician | *Vinader Computación, Encarnación* 2017

- Installed, configured and repaired hardware and software for customers; preventive and corrective maintenance.

Earlier: Cashier, Casa Jorge, Encarnación (5 years): customer service, payments and daily register balancing.

EDUCATION & LANGUAGES

B.Sc. Informatics Engineering | *Universidad Nacional de Itapúa (FIUNI)* Thesis stage, in progress

Technical High School Diploma in Informatics | *Distinguished Student 2016 & 2017* 2015 – 2017

Languages | *Spanish (native), English (C1)*